

regarding the specific skills and contexts required for AI integration. Additionally, the GENOE index considers the complex interplay of tasks within occupations and integrating the ethical and social contexts organically.

Another contribution of our paper is the novel use of large language models to generate synthetic assessments that aim to replicate and extend expert assessments. Many studies rely on expert surveys or interviews to build analytical tools like indexes, including in the labor automation area (Frey and Osborne, 2017; Felten *et al.*, 2021; Tolan *et al.*, 2021). Synthetic AI surveys offer several advantages. They can efficiently process and synthesize vast amounts of information from diverse sources very rapidly, enabling a larger volume of questions to be addressed in a shorter time frame. Additionally, synthetic AI surveys can be easily scaled to include extensive sample sizes or repeated iterations, allowing greater granularity of the collected data.

While recent research has begun to explore the use of AI in workforce analysis, our methodology offers a novel approach in this evolving field. For example, Eloundou *et al.* (2024) combined human expert insights with AI-generated assessments to evaluate AI’s potential for enhancing task efficiency across occupations. Our work is a step forward in this approach by exclusively and systematically using ChatGPT to build the entire index through multiple iterations and varying model parameters.

To validate the accuracy of this methodology, we replicated the well-known exposure index of Felten *et al.* (2021), replacing mTurk surveys with synthetic AI surveys. Our approach successfully replicated Felten’s AI Occupational Exposure Index (AIOE) with a correlation of 0.99, suggesting our methodology is robust and reliable for these types of tasks and providing evidence of another area where AI-driven automation is highly feasible.

The remainder of this paper is organized as follows. After this introduction, Section 2 details our data sources and methodology used to build the GENOE index. Section 3 presents the GENOE index, including its key findings and implications. This section also includes several robustness checks to validate our results and replicates other human-like expert assessment using our novel synthetic survey methodology. Section 4 explores how the index can be used in practical applications, analyzing occupational exposure in the U.S. and Mexico labor market. Finally, Section 5 discusses some considerations and limitations of our research, while Section 6 concludes with implications and directions for future research.

2 DATA AND METHODOLOGY

We draw data from the O*NET database, which contains detailed descriptions and attributes for 1,016 occupations in the U.S. labor market under the O*NET-SOC 2019 classification. Occupational characterizations are built and continuously updated from surveys and input from multiple sources, including job analysts, incumbent workers, and industry experts. O*NET attributes—such as abilities, skills, knowledge, and tasks—are divided into six major domains: Worker Characteristics, Worker Requirements, Experience Requirements, Occupational Requirements, Work-

force Characteristics, and Occupation-Specific Information.

Using O*NET’s task module, which contains 18,156 specific tasks, we describe each occupation as a vector of tasks typically performed by that role. We focus on tasks rather than skills or abilities, following the extensive “task approach” literature (Autor *et al.*, 2003, 2006; Acemoglu and Autor, 2011; Autor, 2013). This framework suggests that technological advancements displace or reinstate specific tasks within occupations rather than entire jobs or skills, as neither skills nor entire jobs directly translate into output (Autor, 2013).

To measure occupational exposure to AI, we employed a novel methodology using large language models (LLMs), specifically the GPT-4o API from OpenAI. Our approach, which we term “synthetic AI surveys,” involves asking these advanced models to estimate the likelihood of AI and its developments replacing each occupation based on its characteristic tasks.

Our synthetic AI survey methodology offers several key advantages in assessing occupational exposure to AI. First, the scalability of AI models allows for rapid processing of vast amounts of information, enabling a comprehensive analysis of all occupations in the database. This scalability ensures that no occupation is overlooked, providing a complete picture of potential AI impact across the labor market. Second, the use of a standardized AI model guarantees consistency in evaluation criteria across all occupations, eliminating potential human biases and ensuring comparability of results. Third, large language models like GPT-4o are trained on recent data, potentially incorporating the latest understanding of AI capabilities and trends, which keeps our analysis current and forward-looking. Lastly, these advanced models offer a more holistic view of tasks, occupations and technological innovations, capable of capturing interactions between tasks and considering broader contextual factors in their assessments.

Key features of our survey methodology include the following:

Independent Sessions: Each survey was conducted in a new session for every occupation, preventing carryover effects from previous responses. This ensures that each occupation is evaluated independently, based solely on its own characteristics.

Multiple Iterations: The process was repeated 100 times to test the accuracy and consistency of the estimates. This multiple-iteration approach allows us to account for potential variability in the AI model’s responses and provides a measure of confidence in our results.

Varied Creativity: The temperature parameter, which controls the randomness and creativity in the model’s output, was randomly set between 0 and 2 for each iteration. This introduces controlled variability in the model’s responses, potentially capturing a wider range of perspectives on each occupation’s AI exposure.

Time Horizons: We considered three exposure horizons: one year, five years, and 10 years. This allows us to capture both immediate and longer-term potential impacts of AI on occupations, recognizing that AI capabilities and their integration into various fields may evolve over time.

The GENOE index for each horizon was calculated as the average value across the 100 sur-